How To Design LED Drivers For Streetlights

This article aims to make design challenges for LED streetlights easier with the help of an example

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In many consumer lighting applications, the cost of existing technology—especially light bulbs or fluorescent lamps—is so low that many of the benefits of LED lighting cannot offset the increase in start-up costs. Street lighting differs significantly because the long life and high degree of control provided by semiconductor lighting is associated with the end user—the government—who evaluates the cost of ownership along with the initial cost.

This is where high-quality LEDs paired with good thermal management and robust drive electronics have real value. This article proposes an electric drive solution that balances higher initial costs with long life—at least as long as of the LEDs' themselves.

DC bus voltages

A hundred or more one-watt LEDs are often enough to produce thousands of lumens needed for a streetlamp. One way to power 100 LEDs would be to string them in one series. This ensures the same current through each LED and, since the light output is proportional to the current, it is the best way to guarantee the same light output from each device.

The problem is that a DC voltage can easily be 400V. Such a voltage can be lethal and also requires large and expensive components.

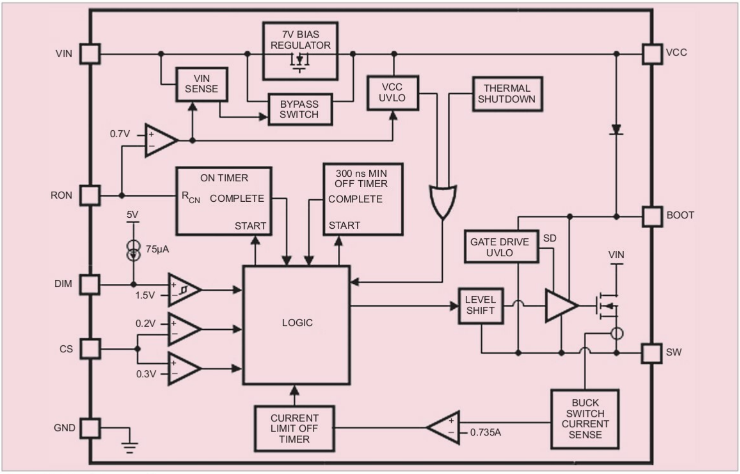

Fig. 1: LED driver fbd_snvs450f